

Topic:

Managing undo data

What is undo?

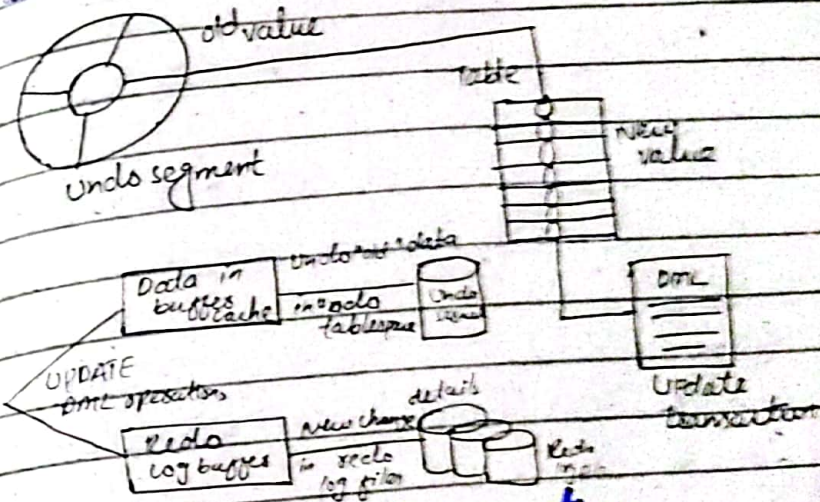
- ⇒ UNDO means to reverse the action of an earlier action.
- ⇒ Oracle records all data that is about to change as undo. The information allows Oracle to undo changes in case of a rollback.

UNDO data:

- ⇒ Undo data is:
 - A copy of original, premodified data.
 - Captured for every transaction that changes data.
 - Retained at least until transaction is ended.
- ⇒ Used to support:
 - Rollback Operations
 - Read-Consistent and flashback queries
 - Recovery from failed transactions

Transactions and Undo data

- Each transaction is assigned to only one undo segment.
- An undo segment can service more than one transaction at a time.



Storing Undo information:

- ⇒ Undo information is stored in undo segments, which are, in turn, stored in an undo tablespace. Undo tablespaces:
 - Are used only for undo segments.
 - Have special recovery transaction
 - May be associated with only a single instance.

- Require that only one of them be the current writable undo tablespace for a given instance at any given time.

UNDO data VS Redo data

Undo	Redo
<u>Record of:-</u>	
◦ How to undo a change.	◦ How to reproduce a change.
<u>Used for:-</u>	
◦ Rollback, read consistency, flashback	◦ Rolling forward database changes
<u>Stored in:-</u>	
◦ Undo segments	◦ Redo log files
<u>Protects against:-</u>	
◦ Inconsistent reads in multiuser systems.	◦ Data loss

Monitoring Undo

⇒ Undo usually requires little management. The areas to monitor include:

- Free space is an undo tablespace
- "Snapshot too old" errors.

Administering Undo

⇒ Administering of undo should include preventing:

- Space errors in an undo tablespace.
 - size the undo tablespace properly.
 - Ensure that large transactions commit periodically.

- "Snapshot too old" errors:

- Configure an appropriate undo retention interval.
- size the undo tablespace properly.
- Consider guaranteeing undo retention.

⇒ Use automatic undo management:

UNDO-MANAGEMENT = AUTO

UNDO-TABLESPACE = UNDOTBS1

Configuring Undo Retention

⇒ UNDO_RETENTION Specifies (in seconds) how long already committed undo information is to be retained.

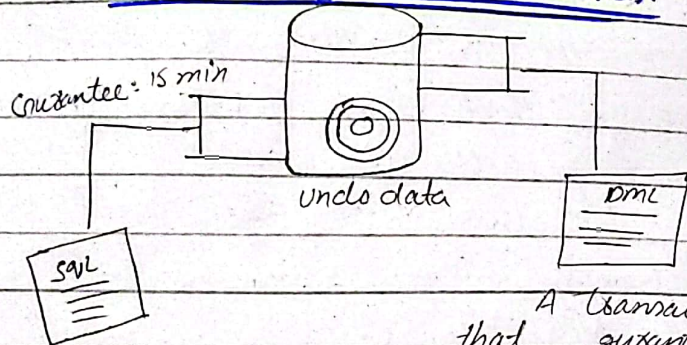
⇒ The undo retention period determines how long undo data is kept before being overwritten.

⇒ The only time must set this parameter is when:

- o The undo tablespace has the AUTOEXTEND option enabled.
- o Want to set undo retention for LOBs

o Want to guarantee retention

Guaranteeing undo retention:



SELECT statement

running is min
or less are always
satisfied

A transaction
that guarantees
more undo than
what there is space
for will fail.

Syntax:

o The syntax to change an undo tablespace to guarantee retention:

⇒ ALTER TABLESPACE undotbs1

RETENTION GUARANTEE;

Sizing the undo Tablespace

o Undo tablespace must be sized so that they can contain the original information for all transactions.

o Datafile belonging to an undo tablespace can automatically extend when they run out of free space.

o Disabling automatic extension in an undo tablespace's data files prevent a single user from inadvertently consuming large amounts of disk space by neglecting to commit transactions.

UNDO advisor:

⇒ The undo advisor accessed through the undo management properties page.

⇒ it provides an estimate of the undo tablespace size required to satisfy a given undo retention.

Q. What is undo data statistics?

◦ it refers to the collected metrics and information that describe the usage and behavior of undo data within the database.

Key aspects of undo data statistics include:

- Usage metrics
 - Transaction information
 - Extent Counts
 - Performance data
 - Retention metrics
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Source: Oracle